# Demo: RaceCert: Exact Certificates for Two-Path Races

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## Abstract

Mobile transports increasingly race packets over paths whose clocks and observations do not share a clean time base. With clock-rate uncertainty, timestamp error, staggered sends, and censored arrivals, one trace can support incompatible path decisions. RaceCert turns that ambiguity into an exact, replayable certificate: it either forces A, proves neutrality within a policy margin, forces B, or supplies a complete feasible counterworld for each remaining outcome. The offline demo combines an exact-rational producer, separately implemented JavaScript and Python verifiers, and a three-socket UDP capture path. Attendees vary timing and policy budgets, trace packet events to decision inequalities, inspect constructive counterworlds, alter and rehash certificate fields, and watch both verifiers reject the changed claim. Six byte-stable fixtures and five live impairment profiles support repeatable interaction; a retained 512-race campaign exposes every input, certificate, aggregate, and digest. RaceCert makes uncertainty in multipath measurement visible as evidence that can be inspected, challenged, and replayed.

## CCS Concepts

• **Networks** → **Network measurement**.

## Keywords

multipath measurement, exact certificates, clock uncertainty, censored observations

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## 1 A path race has more than one clock

Path selection alone can materially change delay, loss, and bandwidth [9]. Multipath TCP exposes concurrent paths to a single transport [12]. On wireless links, interface heterogeneity also reshapes transfer latency [2]. BLEST responds to the resulting head-of-line blocking at the scheduler [5]. STORM extends multipath scheduling to unstable mobile conditions [6]. TANGO shows how route control can widen the set of usable Internet paths [1]. Optimization, however, still depends on a more basic measurement question: *which policy outcome is forced by this observed race and its declared uncertainty budgets?*

For path $i \in \{A, B\}$, let real send and receive times be $s_i$ and $r_i = s_i + d_i$. Endpoint offsets cancel in

$$d_A - d_B = (r_A - r_B) - (s_A - s_B). \tag{1}$$

Packet-transit calibration shows why separate clocks require explicit treatment [8]. Clock skew distorts network-delay measurements [7]. Internet software clocks also exhibit synchronization error [11]. Timestamp reads contribute bounded measurement error [4]. If endpoint $e$ records difference $x$, its rate is $q \in [1 - \epsilon_e, 1 + \epsilon_e]$ for $0 \le \epsilon_e < 1$, and each read error is bounded by $\eta_e$, the exact elapsed-time hull is

$$Q_e(x) = \text{hull}\left\{ \frac{x - \delta}{q} : |\delta| \le 2\eta_e,\ |q - 1| \le \epsilon_e \right\}. \tag{2}$$

RaceCert does not infer these budgets. On physical devices, $\epsilon_e$ can be set from the maximum rate error observed over calibration windows at least as long as the race, rounded outward with a guard margin; $\eta_e$ can combine timestamp resolution with maximum observed read jitter, again rounded outward. A changed clock or timestamp path requires recalibration or a wider bound. The verifier cannot detect a bound that is too small: the certificate may remain internally valid even though the measurement claim is not. The interface therefore shows every declared bound beside the observation and result.

Relative Delay Estimation compares endpoint-local differences [10]. Differentiated pacing applies the same principle to inter-path one-way delay [3]. For paired arrivals, RaceCert computes $I = Q_r(R_A - R_B) - Q_s(S_A - S_B)$. If only A arrives before receiver close $C$, event order gives $r_A - r_B \le r_A - C$ and a sound one-sided interval; only B is symmetric, while a trace with no arrival remains unbounded. For example, a handset can race the same request over Wi-Fi (A) and cellular

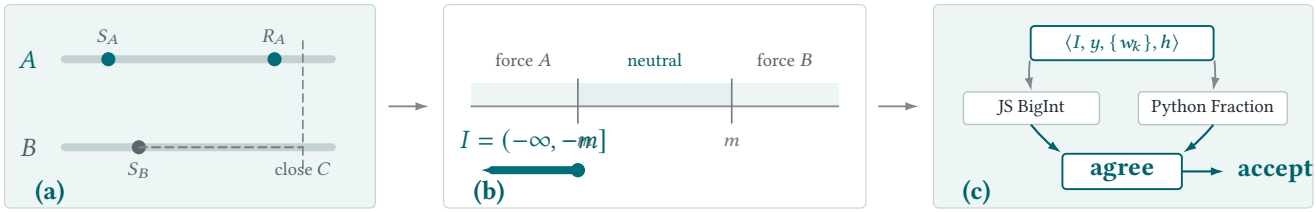



**Figure 1: A packet race becomes a replayable claim: A arrives and B is censored at close $C$ (a); at the exact boundary $u = -m$, $I = (-\infty, -m]$ intersects force A and neutral (b); a certificate carries the outcome and every feasible counterworld to two separately implemented verifiers (c).**

(B) to a common receiver. Sender-local sends, receiver-local arrivals or close, and conservative endpoint bounds map directly to the schema above. The booth uses loopback sockets for repeatability: the setup exercises the full certificate path without claiming to reproduce radio dynamics.

With policy margin $m \geq 0$, every $z \in I$ must satisfy $z < -m$ to force A, lie in $[-m, m]$ to prove neutral, or satisfy $z > m$ to force B. Intersections with several regions produce an ambiguous certificate. For each feasible region, the certificate supplies sender and receiver rates, clock offsets, read errors, real sends, receives, close time, delays, delay difference, and completion waits. Verification checks the affine clock equations, budgets, event order, nonnegative delays, policy region, and interval membership.

For an exact boundary example, let $m = 1$ ms and consider an A-only trace with $I = (-\infty, -1\,\text{ms}]$. Then $z = -2$ ms is a force-A witness, while the feasible endpoint $z = -1$ ms is neutral. The certificate is therefore ambiguous between A and neutral, not force-A. Either witness proves only that its region is feasible; completeness comes from deriving $I$ over all allowed rates and read errors and testing every policy region it intersects.

Across $N$ races, one fixed denominator retains forced, neutral, ambiguous, and censored observations. A path majority requires more than $N/2$ force certificates for that path. Unresolved races therefore stay visible in both the arithmetic and the interface.

## 2 The certificate is the demo

All schema integers are decimal strings, and semantic arithmetic uses reduced exact rationals. The producer emits a canonical JSON digest with each certificate. The JavaScript verifier implements its own rational, interval, witness, canonicalization, and SHA-256 logic; the separately written Python verifier uses Python's exact rational type. Each starts from the original observations and reconstructs every derived field before accepting an individual race or a batch.

The booth keeps acquisition self-contained by binding sender, impairment relay, and receiver sockets to the loopback interface. Relay timers and drops create a UDP log. Captured monotonic times are projected into logical sender and receiver observations rather than replaced by a preset result. Five selectable profiles induce asymmetric delay, B-before-A reordering, selective drop, deadline-late delivery, and a drop-both prelude followed by recovery. Six separately labelled frozen fixtures reproduce byte-identical certificates and make every interaction resettable. An external device trace uses the same schema: the sender contributes $S_A$ and $S_B$; the receiver contributes $R_A$, $R_B$, or close $C$; and the run records the four endpoint budgets. The producer and both verifiers then follow the same path as in the loopback demo. In the present demo, loopback logs supply these fields; a phone deployment would supply them after calibration.

The audience follows one visible chain. Selecting a trace highlights observed timestamps, uncertainty budgets, exact interval endpoints, and the policy regions touched by that interval. Changing the clock-rate, timestamp-error, or policy-margin controls can move the same observation between decisive and ambiguous outcomes. A forced result opens the inequalities excluding the other regions; an ambiguous result opens one complete counterworld per feasible region. One frozen fixture exposes the censored $u = -m$ boundary above, including its neutral witness. A live profile reveals acquisition events, while another fixture isolates semantic replay. Editing an outcome or witness and recomputing a valid SHA-256 still fails both verifiers: rehashing authenticates the modified bytes, but the altered claim no longer satisfies the observation.

The five-minute presentation follows one claim from capture to replay. It first holds the observation fixed while widening one uncertainty budget, then opens the censored boundary fixture, mutates its witness, and ends with the 512-race batch. One command launches the server and browser with bounded fallback across busy localhost ports.

Displayed capture events and declared uncertainty budgets anchor each result. Both verifiers then recompute the hull, policy relation, counterworld constraints, and aggregation. Capture provenance remains an input declaration; every certificate-derived claim is replayable.

**Table 1: Replayed evidence behind the live demo.**

| Check | Reproduced result |
|---|---|
| Frozen fixtures | 6 byte-stable; 5 full witnesses checked |
| Live UDP profiles | **5/5 pass** both verifiers |
| Rehashed attacks | outcome and witness: **both reject** |
| Seeded campaign | 512 races: A 176; neutral 12; B 191; ambiguous 133 |
| Fixed denominator | $N = 512$; no forced majority; **both pass** |

Each certificate has two complementary readings. Geometrically, $I$ is the complete feasible set of delay differences induced by the recorded clocks and error budgets; its overlap with the policy partition determines the result. Constructively, every ambiguous branch carries concrete clocks, errors, sends, receives, and waits that realize that branch. These witnesses are feasible counterexamples, not estimates of the hidden clocks that actually occurred. The interface keeps the two views synchronized: selecting a policy region highlights its witness, and selecting a witness traces each value back to the observed event or declared bound that permits it. This turns ambiguity into an object to inspect rather than a label to accept.

## 3 Evidence that can be broken and rebuilt

The fixed-seed campaign uses a documented 64-bit xorshift generator and retains every generated input with its certificate. Its 512 races yield 176 force-A, 12 neutral, 191 force-B, and 133 ambiguous certificates. Neither forced count exceeds 256, so both verifiers derive the same no-majority result. The campaign exercises the complete certificate pipeline across a broad deterministic input set and leaves every aggregate linked to its constituent races.

Automated tests cover forced-A, forced-B, neutral, and no-majority aggregation; five complete ambiguity witnesses; capture-specific delay, reorder, drop, deadline, and recovery events; and two rehashed semantic attacks. The JavaScript and Python implementations share wire data and test cases while recomputing arithmetic through different numeric code. Across five complete test runs, every live profile passes and the B-before-A relay order is reproduced in 100 stress captures. Interface checks at 1280×720 and 390×844 report no horizontal overflow or console error.

These results establish exact replay, cross-language verification, and the five loopback acquisition profiles. They do not validate the bounds chosen for a physical phone trace or compare radio-path performance; those remain deployment inputs. Once supplied, the same producer and verifiers check the resulting claim.

RaceCert leaves every race inspectable: decisive traces carry inequalities, ambiguous traces carry counterworlds, censored traces stay in the denominator, and semantic edits trigger reproducible rejection. Path uncertainty becomes evidence an audience can challenge and replay.

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